

Who is building the engineered world

WHO IS BUILDING THE ENGINEERED WORLD, AND HOW TO ARGUE WITH THEM

A plain guide to the genetic engineering industry, the laws that let it work, and where an outsider can get a grip

BEFORE ANYTHING ELSE

This guide is not about stopping a particular field trial. There is a separate guide for that. This one is about the thing behind the field trials: an industry of a few very large companies, a set of laws that were mostly written before the technology existed, and a decision-making system that almost nobody outside it has ever looked at.

It assumes you know nothing about the subject. It starts from what genetic engineering actually is, goes through where the industry came from, how it works today, and where it is heading, and then gets to what you can do — which is the point of the whole document.

Two things about the position it takes, stated at the front so you can decide whether to trust the rest.

The objection here is not to the organisms. A gene-edited plant is not wicked. The objection is to the machine around it: a handful of firms that own the seed and the chemical the seed is built to survive, a public that pays for much of the research and is not asked about the results, safety files written by the applicant, and rules being rewritten faster than anyone outside the room can follow. That is a governance problem. It would be a governance problem if the products were flawless.

The scale is not ordinary. Cheap editing, cheap DNA synthesis, and software that designs sequences have put a set of tools in reach that operate on anything alive: crops, livestock, insects, microbes, forests, and — with companies now funded and recruiting for it — human embryos. Some of what is being built is intended to spread on its own and cannot be recalled. The people deciding whether that happens are, in most countries, a committee you have never heard of, working from a file the company wrote.

You do not need a science degree to argue with any of that. You need to know where the decisions are made and when they are open. Both are public.

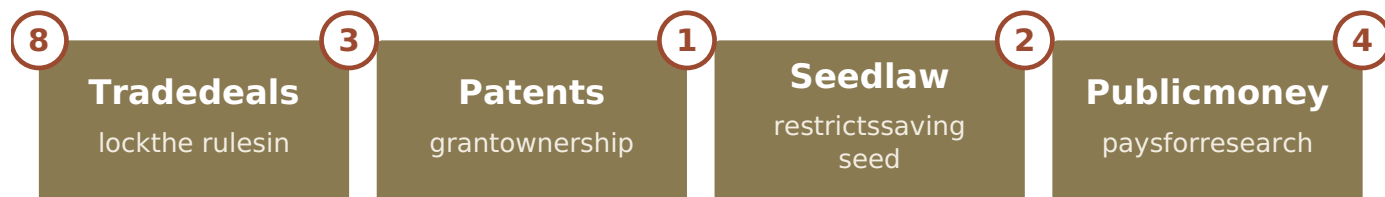
THE MACHINE, ON ONE PAGE

Before the history, the picture. Everything in this guide is a close-up of one part of the diagram below, and the last figure comes back to it to show where the handles are.

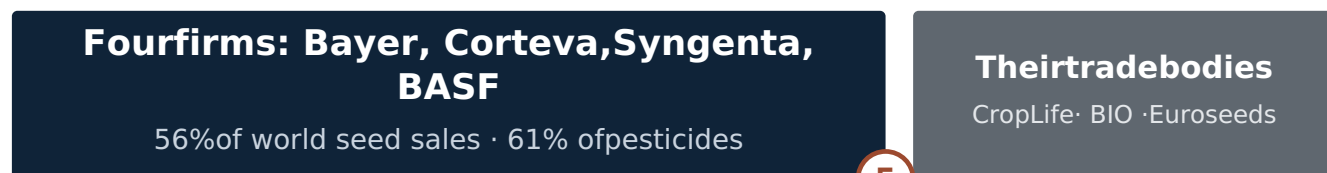
The machine, on one page

Money and law at the top, consequences at the bottom, and ten places where an outsider can push.

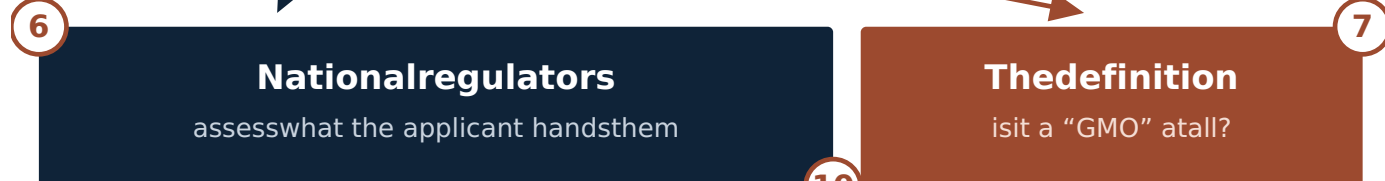
WHAT FEEDS IT



WHY RUNS IT



THE RULE THEY SHAPE



WHAT GETS RELEASED



WHO CARRIES THE RESULT



The numbered circles are the ten places an outsider can get hold of. They are listed under the figure.

Figure 1. The machine, on one page. **Top:** what feeds it — trade deals, patents, seed law and public money, none of which is a biosafety body. **Middle:** four firms, their trade associations, and the regulators whose files those firms write. **The rust box** is the question that decides whether any of the rest applies. **Bottom:** what gets released, and who lives with it. The numbered circles mark the ten openings listed below.

The ten numbered circles are the openings this guide is about:

1. **Oppose the patent.** Patent offices run public objection procedures.
2. **Fight the seed law.** Who may legally save and exchange seed is set by national law, not by biology.
3. **Read the trade deal.** Trade agreements lock rules in for decades and are negotiated with almost no public participation.
4. **Follow the public money.** A large share of the research is publicly funded, and public funders answer to legislatures.
5. **Ask who did the safety study.** In most systems the applicant supplies the evidence about its own product.
6. **Comment on the rule.** Rule changes decide thousands of future products at once, on a public docket almost nobody reads.
7. **Defend the definition.** Whether an organism legally counts as engineered decides whether any of the rest applies to it.

8. **Go to the treaty meetings.** They accept written submissions from anyone.
9. **Reach the buyers.** Supermarkets, grain traders and food companies have no deadlines and no legal obligation to say yes.
10. **Litigate on process.** Courts rarely rule on whether something is safe. They rule on whether the agency followed its own rules.

Four things are worth noticing in the diagram before you read on.

The money and the law sit above the science, not below it. Patent offices, seed law, trade agreements and research funders shape what gets built. None of them is a biosafety body, and none of them assesses risk.

The regulator in the middle is not the top of the system. It is closer to the middle of it, and it works largely from what the applicant hands over.

The box on the right of the rules band decides whether the rest of the diagram applies at all. If a government rules that a gene-edited organism is not legally a GMO, it never enters the system. That single question is now the main battleground worldwide.

The bottom row has no seat in the top row. Farmers, ecosystems, eaters and the people who inherit the results are where the consequences land. Everything in the “what you can do” half of this guide is a way of building a connection from the bottom row to the top one.

PART ONE: WHAT THE FIELD ACTUALLY IS

The techniques, from oldest to newest

You will meet these terms constantly. They are simpler than they sound.

Selective breeding is what farmers have done for ten thousand years: keep the seed from the best plants. Slow, no laboratory.

Mutagenesis means deliberately damaging DNA at random with chemicals or radiation, growing out thousands of plants, and keeping any useful accidents. It has been used since the 1930s. In most countries it is legally exempt from GMO rules, which matters more than it sounds — the exemption is the precedent everyone now argues from.

Transgenesis is moving a gene from one species into another. A soil bacterium’s insecticide gene into maize; a bacterial gene that makes a plant survive a weedkiller. This is what “GMO” originally meant, and it still accounts for nearly all the engineered crop area planted worldwide.

Gene editing — CRISPR is the best-known tool — makes a targeted change to an organism’s own DNA without necessarily adding anything foreign. It is cheaper and faster than transgenesis by a wide margin. Whether an edited organism should count as a GMO is the central legal argument of the last decade, and it is the argument the industry is winning.

Gene drives are engineered systems designed to spread a trait through a wild population much faster than ordinary inheritance would. Normal genes pass to about half the offspring; a drive is built to pass to nearly all of them. The point is that it spreads on its own, which is also the reason there is no practical way to recall one.

Synthetic biology is building genetic sequences from scratch rather than moving existing ones, increasingly with software doing the design.

There is a separate branch that people often do not think of as genetic engineering at all: **assisted reproduction**, cloning, and embryo screening and selection. Technically these are not all genetic modification. They belong in the same conversation because they are the same industrial machinery pointed at human reproduction, run by companies, with the same pattern of self-governance.

What is actually out there now

Approximate, and worth knowing so you neither overstate nor understate:

- Engineered crops cover roughly **216 million hectares** across about **30 countries**. Four crops — soy, maize, cotton, canola — are nearly all of it.
- Three countries account for the large majority of that area: the **United States, Brazil and Argentina**, with **Canada** and now **China** growing fast. China’s engineered maize and soy reached roughly **three million hectares** in 2025, four to five times the 2024 area.

- The most common engineered trait by area is **herbicide tolerance** — the plant survives a weedkiller so the field can be sprayed. Insect resistance is second. Traits that consumers would notice, like better nutrition, are a small fraction.
- Gene-edited crops are approved in a growing number of countries and, in many of them, are **not tracked as engineered at all**, so nobody can tell you how much is planted.
- Engineered animals — mainly pigs and cattle — have been cleared in several countries. Engineered salmon has been on sale.
- Engineered microbes make a large and mostly unlabelled share of food ingredients: enzymes, vitamins, flavourings, and now proteins made by fermentation.
- No gene drive organism has been released anywhere. Developers have said they are aiming at field trials around **2030**.
- Heritable editing of human embryos to make babies is banned or restricted in most countries. At least three startups are funded and working on it.

What is genuinely settled, and what is not

A guide that pretends everything is contested is as useless as one that pretends nothing is. Where the lines actually fall:

Reasonably settled. The large scientific and food-safety bodies that have reviewed the engineered foods currently approved have concluded they are not more dangerous to eat than the conventional equivalents. You can argue about who funded those reviews and how they were done. But if you build a campaign on a health scare, the other side answers it in a paragraph and then uses it to dismiss everything else you said. It is the single most common way opposition campaigns lose credibility, and it is avoidable.

Genuinely contested, with evidence on both sides. Whether engineered crops have increased or decreased total pesticide use, and over what timescale. Whether the yield gains are real once you compare like with like. Whether insect resistance to Bt traits is being managed or merely delayed.

Not seriously contested, and where your case lives. Herbicide-tolerant crops are the largest category by area and they exist to be sprayed. Weeds have evolved resistance and spraying has escalated. Engineered traits move: the GM Contamination Register logged 396 incidents of contamination, illegal planting and negative agricultural side effects across 63 countries between 1997 and 2013. Almost no country has a working liability rule for what happens when they do. Almost no country funds the post-approval monitoring its own approvals promise. Four firms hold 56% of the world seed market. And for anything designed to spread on its own, no recall exists.

Those last points are all about governance and ownership, not biology. They are documented, checkable, and they are what the rest of this guide is built on.

PART TWO: WHERE IT CAME FROM

The industry is not the product of a single decision. It is the product of about six, taken between 1975 and 1986, most of which had nothing to do with biology. Knowing them matters because every one of those decisions is still operating, and several are still open to challenge.

Where it came from

Three turns that set the rules still in force today.

THE DISCOVERY

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|-------------|--|
| 1953 | The structure of DNA is published. |
| 1973 | Cohen and Boyer move a gene from one organism into another. The technique works. |
| 1975 | At Asilomar, scientists write their own safety rules and agree to police themselves. No law is passed. The habit of self-governance starts here and never really ends. |

LIFE BECOMES PROPERTY

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|-------------|---|
| 1980 | Diamond v. Chakrabarty: the US Supreme Court holds that a living organism can be patented. |
| 1980 | The Bayh-Dole Act lets universities patent research they receive from the public, and license it to companies. |
| 1982 | The first recombinant drug, human insulin, is approved. Medicine and agriculture part company in the public mind. |

THE RULES ARE SET

- | | |
|-------------|---|
| 1986 | The US Coordinated Framework. Rather than write a new law, agencies stretch statutes meant for pesticides, food and plant pests. Forty years later this is still the basis. |
| 1991 | UPOV 91 narrows the farmer's right to save and exchange seed. |
| 1994 | The Flavr Savr tomato reaches shops. |
| 1996 | Herbicide-tolerant soy is planted commercially. Adoption is faster than that of almost any farm technology before it. |

Figure 2. Three turns that built the system. None of them was a decision about biology. Asilomar set the habit of self-governance; 1980 made organisms property and let public research be privatised; 1986 decided not to write a law at all.

1975: the field agrees to police itself

In 1975, at Asilomar in California, the scientists who had just learned how to move genes between organisms met to decide what to do about it. They agreed categories of risk and laboratory precautions, and they resumed work.

Asilomar is usually told as a story about responsible science, and in its own terms it was. It is worth noticing what else it did. It established that this field's safety rules would be written by its practitioners rather than by legislatures, and that the question would be framed as laboratory containment rather than as who should decide. Fifty years later, the pattern holds:

the 2019 call for a moratorium on heritable human genome editing was issued by scientists, is not law anywhere it matters, and startups are now proceeding against it.

When you hear that a field is “self-regulating” or has agreed a “voluntary moratorium,” this is the lineage. A moratorium with no legal force is a statement of intent by the people who would have to break it.

1980: living organisms become property

Two things happened in the United States in 1980 that shaped the industry more than any biosafety rule since.

In *Diamond v. Chakrabarty*, the US Supreme Court held that a living, human-modified organism could be patented. And the Bayh-Dole Act let universities take out patents on research the public had paid for, and license them exclusively to companies.

Together these made the modern arrangement possible: public money funds the science, a patent turns the result into private property, and the licence turns it into a product. Nearly every argument about seed ownership, farmer dependence, and who profits from publicly funded research traces back to this pair.

1986: the decision not to write a law

Faced with a new technology, the United States chose not to pass a new statute. The 1986 Coordinated Framework declared that existing laws were sufficient, and split responsibility between three agencies using powers written for other purposes: the agriculture department under a law about plant *pests*, the environment agency under a law about *pesticides*, and the food agency under a law about *food additives*.

This is not a historical curiosity. It is the live basis of US oversight today. It explains why the agriculture department’s authority hangs on whether a plant pest was used in development — a question with no relationship to whether the resulting organism poses any risk — and why an organism made without one can fall outside the rules entirely. Most countries that built frameworks later looked at this arrangement when designing their own.

1991: the seed treaty

UPOV — the International Union for the Protection of New Varieties of Plants — is a treaty most people have never heard of that decides something everyone understands: whether a farmer may save seed from this year’s harvest to plant next year.

The 1978 version left that largely alone. The 1991 version narrowed it sharply, making saving, exchanging and selling protected varieties a matter of the breeder’s permission. As of mid-2025, UPOV had **80 members covering 99 countries**, and UPOV 91 compliance is regularly written into trade agreements as a condition.

The consequences on the ground are concrete. Kenya’s seed law, amended in 2012 along UPOV 91 lines, makes uncertified seed exchange an offence. Ghana’s Plant Variety Protection Act of 2020 carries a minimum ten-year prison term for infringing breeders’ rights. Seventeen West African countries are bound to UPOV 91 through their membership of a regional intellectual property organisation, without each having decided it separately.

If you want to understand why farmers’ organisations across Africa, Latin America and Asia treat seed law as the front line rather than biosafety law, that is why. Biosafety law asks whether a product may be sold. Seed law decides whether you may keep your own.

1996 onward: commercialisation, and the world’s answer

Commercial planting began in 1996 and spread faster than almost any farm technology before it. The international response was the **Cartagena Protocol on Biosafety**, adopted in 2000 and in force from 2003, now with **173 parties**. It is the only binding global instrument on this subject. It gives importing countries the right to be notified and to decide before living modified organisms are first sent to them for release, and it created the Biosafety Clearing-House, a free public database of national decisions and authorities.

The United States — the largest producer — is not a party.

Howit gothere

Threemore turns, and the fight the field is innow.

THEWORLDANSWERS

2000	TheCartagena Protocol is adopted; in force from 2003, now173 countries.The United States is not one ofthem.
2001	EUDirective 2001/18 sets the strictest majorframework.
2010	Indiasuspends Bt brinjal after public consultations heldacross thecountry. The moratorium stillstands.

THECOSTCOLLAPSES

2012	CRISPRis published. Editing becomes cheap, fast andavailable toalmost anylaboratory.
2015	Argentinais the first to place gene-edited crops outsideGMO rules.Roughly two dozen countriesfollow.
2018	HeJiankui announces gene-edited babies. He is imprisoned.The fieldagrees a moratorium it has no power toenforce.

THEDEFINITIONFIGHT

2024	AUS court vacates the SECURE rule. India's Supreme Courtsplits onGM mustard and orders a national policy. Mexico loses atrade panelover its maizerestrictions.
2025	Mexicoamends its constitution to prohibit planting GMmaize. BurkinaFaso suspends the engineered-mosquitoprogramme.
2026	EUREgulation 2026/1388 splits gene-edited plants intotwo tiers,with no patent ban. USDA consults on whatreplaces SECURE.The argument is now about definitions, not products.

Figure 3. Three more turns, and the argument the field is in now. Note what changed between 2000 and 2026: the question moved from whether a product is safe to whether it is legally visible.

2012 onward: the cost collapses and the definition fight begins

CRISPR was published in 2012. What changed was not what was possible but what was cheap. Work that needed a well-funded corporate laboratory became accessible to a university department, a startup, and eventually a small company anywhere in the world.

That produced the argument the field is now in. If an edited organism carries no foreign DNA and could in principle have arisen by mutation or conventional breeding, should it be regulated as engineered at all?

Argentina answered no first, in 2015. Chile, Brazil, Colombia, Paraguay, Honduras, Guatemala, Costa Rica and Uruguay followed with near-identical rules. Nigeria, Kenya, Malawi, Rwanda, Ghana, Ethiopia and Burkina Faso have since issued guidance along similar lines. Japan, England, Canada and Australia have each moved in the same direction by different routes. In the **United States, Canada and Australia**, developers may in some circumstances determine their own

regulatory status without submitting anything.

The European Union, historically the strictest, adopted **Regulation (EU) 2026/1388** on new genomic techniques — in force 16 July 2026, applying from 17 July 2028 — which splits gene-edited plants into a lighter-touch tier and a tier that stays closer to the old rules. The European Parliament had voted in February 2024 to ban patents on such plants. The final text agreed in December 2025 does not contain that ban.

This is the most important thing to understand about the present moment. **The argument is no longer mainly about whether particular products are safe. It is about which products are legally visible at all.** A campaign that fights permits while the category empties out is fighting the last decade’s fight.

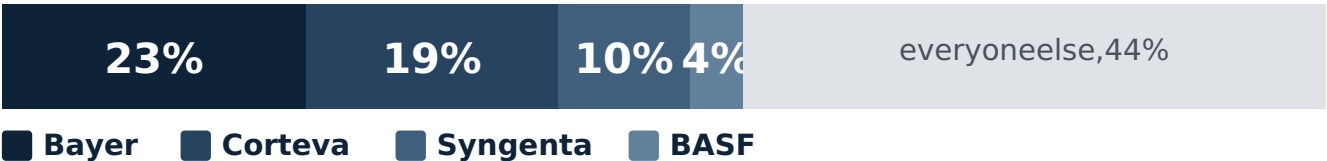
PART THREE: HOW THE MACHINE WORKS NOW

Who owns it

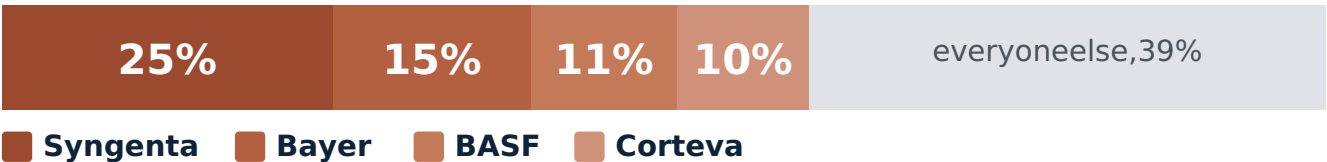
Four companies

Shares of the world market, 2025.

COMMERCIAL SEED — ABOUT \$50 BILLION A YEAR



PESTICIDES



The same four firms sell the seed and the chemical the seed is engineered to survive.

Economists treat a 40% share for the top four firms as the point where market power starts to distort a market. Both of these are above 55%.

Figure 4. Shares of the world market, 2025. The same four firms appear in both bars, in a different order. Read them together: the company selling the seed generally also sells the chemical the seed is engineered to survive.

Four companies — Bayer, Corteva, Syngenta and BASF — hold **56% of the world commercial seed market and 61% of the pesticides market**. Bayer alone has about 23% of seed. Syngenta alone has about 25% of pesticides. The top nine seed firms together hold about 67% of a market worth roughly \$50 billion a year.

Two features of that structure matter more than the raw numbers.

They sell both halves of the pair. The dominant engineered trait by area is tolerance to a herbicide. The company selling the seed generally also sells the herbicide the seed is built to survive. The product is not really the plant; it is the system.

They got here by merging with each other. The present four are what remained after the 2015–2018 wave: Bayer bought Monsanto, Dow and DuPont merged into Corteva, ChemChina bought Syngenta, and BASF picked up the assets competition

authorities forced Bayer to sell. Economists generally treat a 40% combined share for the top four firms as the level where market power starts distorting a market. Both of these markets are above 55%.

Bayer's position also shows the pressure the structure is under. Its share price fell sharply after the Monsanto purchase, largely because of court losses over glyphosate, and it began cutting thousands of jobs in 2023. A campaigner should read that two ways: these companies are not invulnerable, and litigation has been the thing that hurt them most.

Where the money comes from

Not only from the companies. Follow four streams, because each has its own accountability channel and each is a place to ask questions:

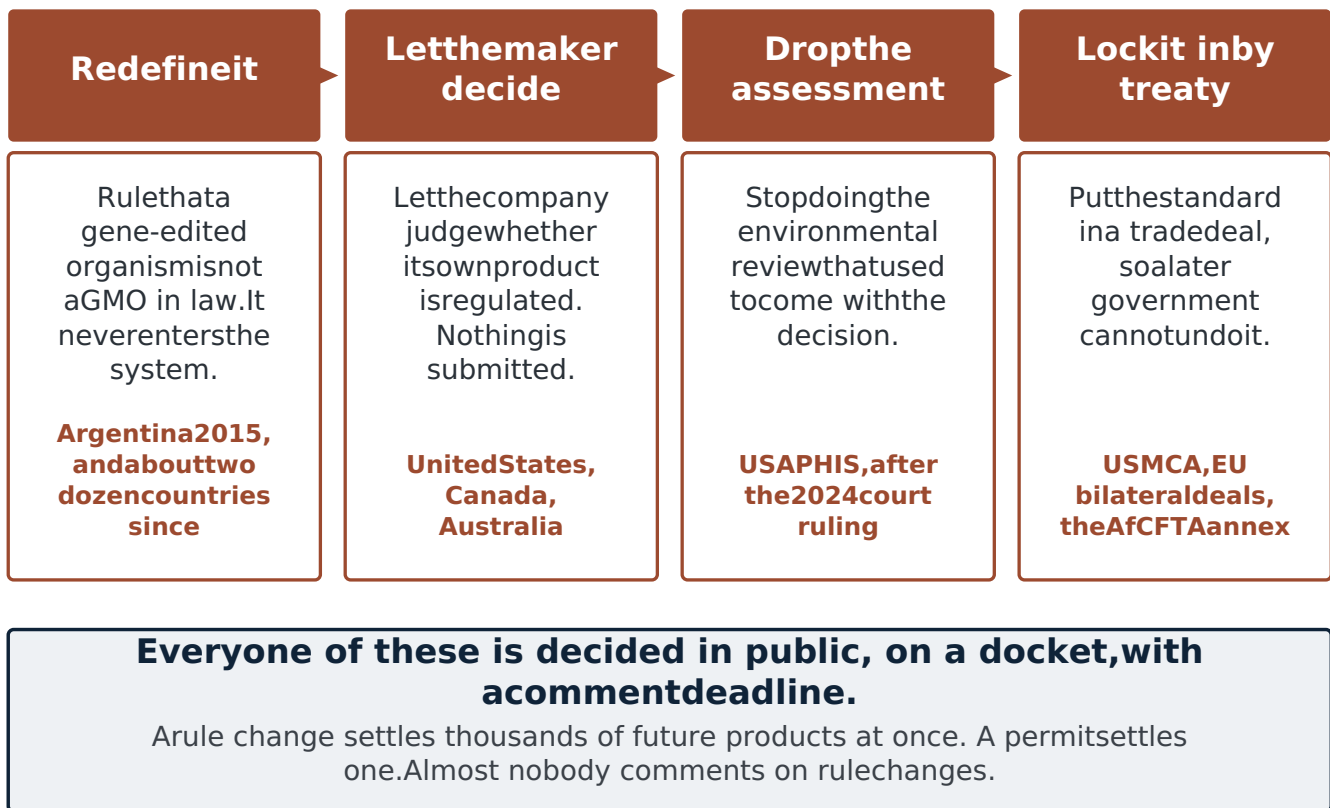
- **Public research funders** — national research councils, agriculture departments, and the international agricultural research network. Most publish searchable grant databases with project descriptions, budgets and timelines. They answer to legislatures and to public auditors.
- **Philanthropy.** Large foundations fund a substantial share of agricultural biotechnology aimed at low-income countries, including the engineered mosquito programmes. Most publish their grants. Most have their own complaints and safeguards procedures, entirely separate from any regulator and far less used.
- **Defence and security budgets,** which fund biological technologies research in several countries.
- **Venture capital,** which is where the newest and least regulated work sits — embryo editing, de-extinction, engineered wildlife.

Public and philanthropic money is the most exposed of the four. It comes with published grant records, reporting obligations, and in many cases a legislature that can be asked why it was spent this way.

How the rules get made, and unmade

How a rule gets loosened

Four moves. Not one of them requires proving that anything is safe.



WHO YOU ARE ACTUALLY ARGUING WITH

- A** Agency rule-writers, on a public docket
- B** Patent examiners and appeal boards
- C** Trade negotiators, who work without a docket
- D** Legislators, who can override all three

Figure 5. Four ways to loosen a rule. Not one requires demonstrating that anything is safe. Three of the four happen on a public docket with a deadline; the fourth happens in a trade negotiation with no docket at all.

This is the part almost nobody outside the industry watches, and it is where the most is decided.

There are four standard ways to loosen a rule, and none of them requires demonstrating that anything is safe:

Redefine it. Rule that the organism is not legally a GMO. Then no assessment, no notification, no comment period and no label applies, because the law never reaches it.

Let the maker decide. Permit the developer to determine its own regulatory status. In the United States, Canada and Australia this is available in some circumstances with nothing submitted to anyone.

Drop the assessment. After the 2024 court decision vacating the SECURE rule, the US agriculture department reverted to its pre-2020 regulations and indicated it would **no longer conduct environmental analyses under the National Environmental Policy Act for deregulation petitions**. The decision-making continues; the environmental review attached to it does not.

Lock it in by treaty. Write the standard into a trade agreement. Then a future government that wants to change it faces a trade dispute rather than a domestic policy debate.

Mexico is the clearest demonstration of the last one. Farmers, scientists and beekeepers filed a collective lawsuit against GM maize in 2013 which froze planting permits for years. A presidential decree later restricted GM corn for tortillas. The United States challenged it under the trade agreement between the United States, Mexico and Canada, and on **20 December 2024** the panel ruled against Mexico. Mexico's answer was to move the question up a level: on **17 March 2025** it published an amendment to Articles 4 and 27 of its own constitution prohibiting the planting of genetically modified maize.

Read that as a two-sided lesson. Trade law is a real constraint that will be used against you. And a rule written into a constitution is much harder to attack through a trade panel than a decree.

Who checks the safety

In most systems, the studies a regulator reviews were produced or paid for by the company seeking the approval. Independent replication is uncommon and rarely funded. Substantial parts of submitted files are frequently withheld as confidential business information — sometimes including the data on which safety claims rest.

This is not a conspiracy allegation. It is the design of the system, it is publicly documented, and it is a legitimate and effective thing to raise. The question “who performed this study, who paid for it, and has anyone independent checked it” is answerable, on the record, and awkward.

Transparency is also going backwards in places. The US agency indicated it will no longer do environmental analyses for deregulation petitions. In China, reporting indicates that from 2025 the agriculture ministry stopped publishing newly approved biosafety certificates. In Canada, the register that does publish gene-edited approvals reveals a structural gap: the OECD unique identifier — the code that lets you track an engineered event across countries — was designed around transgenic insertions. Where there is no inserted transgene, there is nothing for the code to name. The fastest-growing and least-regulated category is also the one that cannot be tracked internationally by the system built for the purpose.

The lobby, and the revolving door

The four firms and their trade associations — CropLife, Euroseeds, BIO — are among the most heavily resourced lobbies in agriculture. Bayer self-declared about €6.5 million of EU lobbying in 2022, roughly €49 million worldwide, plus about €26 million in trade association fees. Corporate Europe Observatory put the core group of firms and associations lobbying against one EU pesticide reduction law at about €15 million of declared spending in a year.

All of this is public, because most jurisdictions have a lobbying register. Committee memberships and previous employers are also public. Checking who sits on an advisory committee and where they worked before is an hour's work and it is often the most quoted thing a small group produces.

Patents: the fight almost nobody joins

Patent law does more to shape this industry than biosafety law, and it has a public objection procedure that almost no citizen group uses.

The European position shows the mechanism. Patents on plants from “essentially biological processes” — ordinary breeding — are prohibited in Europe, a point the EU made explicit in 2017. What is happening instead is that naturally occurring gene variants are patented as inventions, and because a patent on a gene extends to plants bred using it, the prohibition is worked around. The campaign coalition No Patents on Seeds documented patents granted in 2025 claiming natural gene variants in spinach, tomatoes and lettuce, and around 40 further applications of the same kind filed in 2025. Their opposition to one maize patent was rejected by the European Patent Office in November 2025; they have appealed.

The practical consequence lands hardest on small and medium breeders, who cannot pay licence fees or retain patent attorneys, and who abandon breeding projects rather than risk infringement. Opinion polling across Germany, France, Italy, the Netherlands and Poland has found around **80% of the public opposed to patents on seeds** — one of the few positions in this whole area with that kind of majority behind it.

The European Patent Office runs a nine-month opposition window after a patent is granted, open to any person. Most national patent offices have an equivalent. This is the single most under-used lever in the field.

PART FOUR: WHERE IT IS HEADING

Where it is heading

One toolkit, seven directions. The status says what has actually happened, not what is promised.

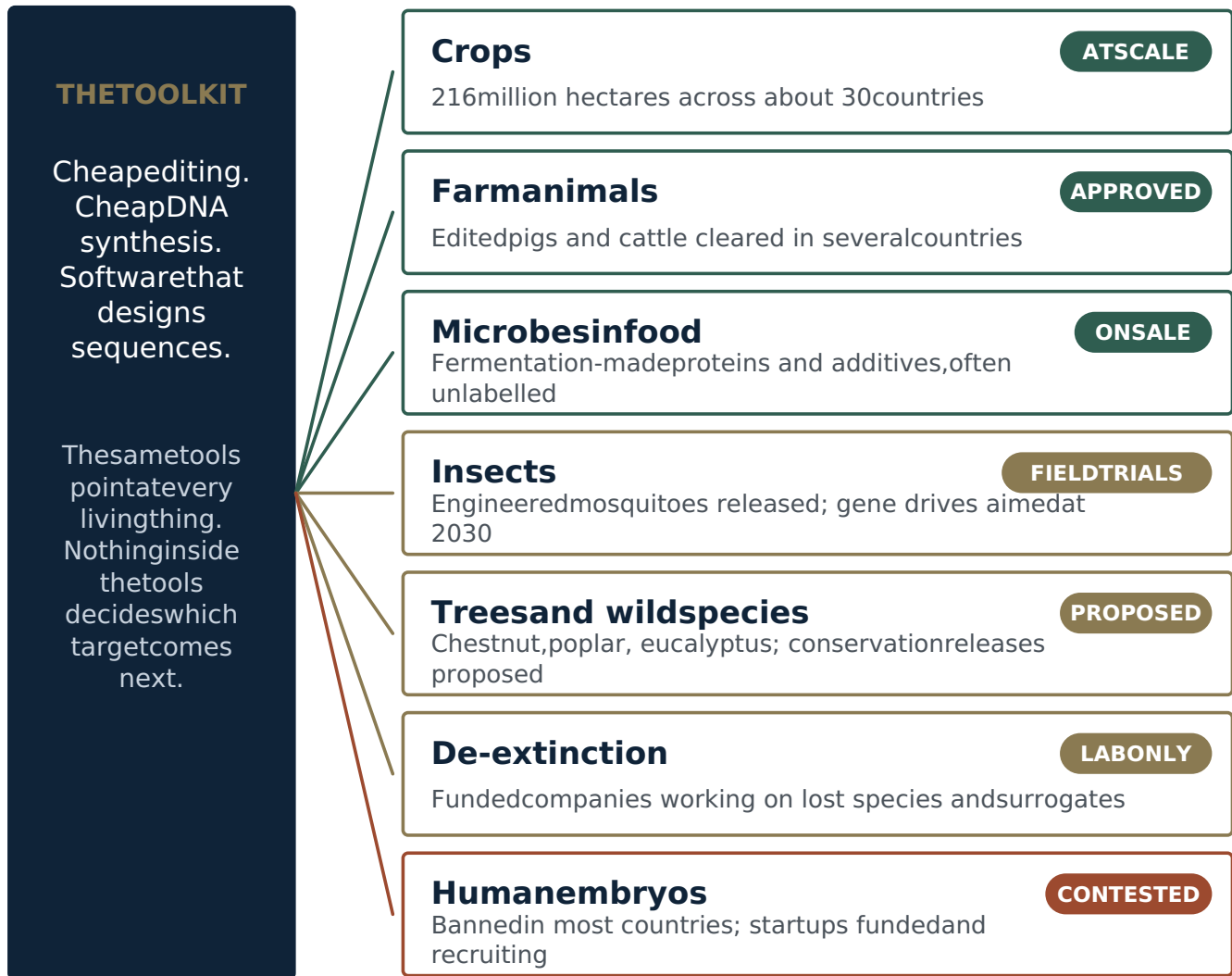


Figure 6. One toolkit, seven directions. The status label says what has actually happened, not what is promised. The three at the bottom are where the governance has least to say and the least public attention.

The toolkit does not care what it is pointed at. Nothing inside cheap editing, cheap synthesis and design software decides which organism comes next. What decides is money and law, which is why those are what this guide is about.

Crops: consolidation, not expansion of the argument

Engineered crop area keeps growing, but the interesting movement is legal, not agronomic. The growth is in countries moving gene-edited crops out of the regulated category. China, which treats agricultural biotechnology as a national security matter, approved its first gene-edited rice for cultivation in December 2024 and expanded engineered maize and soy four to five fold in a single year, to roughly three million hectares.

Farm animals

Gene-edited pigs and cattle have been cleared in several countries. The regulatory questions are the same as for crops, with animal welfare added and, in most places, no labelling requirement at all.

Microbes and fermentation

The least visible category and possibly the largest. Engineered microorganisms produce enzymes, vitamins, flavourings and now proteins used in food. In many jurisdictions, if the final product contains no introduced genetic material, it is not treated as a GM food and is not labelled. Most people eating these products have no way to know.

Insects, and the gene drive question

Engineered mosquitoes have been released in several countries. Those releases so far have used self-limiting designs: the modification is meant to die out.

Gene drives are the other thing. No gene drive organism has been released anywhere. Developers have said publicly they are aiming at field trials in a malaria-endemic African country around **2030**.

Whatever you conclude about the goal — and malaria kills hundreds of thousands of people a year, overwhelmingly children, which is not a small thing to weigh against — the governance question is unusually sharp. A drive is designed to spread through a wild population. It does not respect borders, so a decision taken in one country determines outcomes in its neighbours. There is no established procedure for obtaining consent from populations across a region, and no demonstrated way to recall one.

Burkina Faso is the case to know. After a release at Souroukoudingan on **11 August 2025**, the authorities sealed the Target Malaria programme's facilities on **18 August** and suspended all its activities on **22 August**. The programme was funded from three continents and had national regulatory approvals, environmental sign-off and ethics committee clearance. Objection raised where the release was actually happening ended it anyway.

Trees and wild organisms

Engineered chestnut, poplar and eucalyptus have been developed, and releasing engineered organisms for conservation purposes — to save a species, to remove an invasive one — is now openly proposed. This scrambles the usual political alignments, because the proponents are often conservationists.

De-extinction

Funded companies are working on lost species and their surrogates. Currently laboratory work with heavy publicity. Worth watching mainly because it normalises releasing engineered organisms into wild ecosystems for reasons that are not agricultural.

Human embryos

Editing embryos to make babies is banned or restricted in most countries. In the United States an appropriations restriction has held the line, and the first scientist to do it, in China, was imprisoned.

Three US startups have been reported as pursuing it: **Preventive**, backed by prominent technology investors and said to be focused on preclinical research for now; **Bootstrap Bio**, reported to be interested in enhancing intelligence and to be planning work in Honduras; and **Manhattan Genomics**, since reported as defunct. None has significant staff or facilities. The 2019 scientists' call for a global moratorium remains what it was: a statement by people who are not the ones proceeding.

The mainstream scientific position is not that this is imminent and fine. It is that preclinical safety has not been established and the societal debate has not concluded, and that for nearly every case of preventing a genetic disease, existing embryo screening already works. That leaves enhancement as the actual frontier, which is a different conversation and one that has not been had.

The three things to watch

Definitions. Every jurisdiction that moves gene editing outside its GMO rules removes a whole class of future products from public visibility. This is the fastest-moving front and the one with the least public attention.

Anything designed to spread. Gene drives and self-propagating conservation releases put an irreversible decision in the hands of one national regulator, with cross-border effects and no consent procedure.

Human germline. Small, well-funded, moving, and governed largely by national criminal law and one US budget rider rather than by anything international.

PART FIVE: WHAT YOU CAN ACTUALLY DO

Where the levers attach

The same machine. Nine things an outsider can actually get hold of.

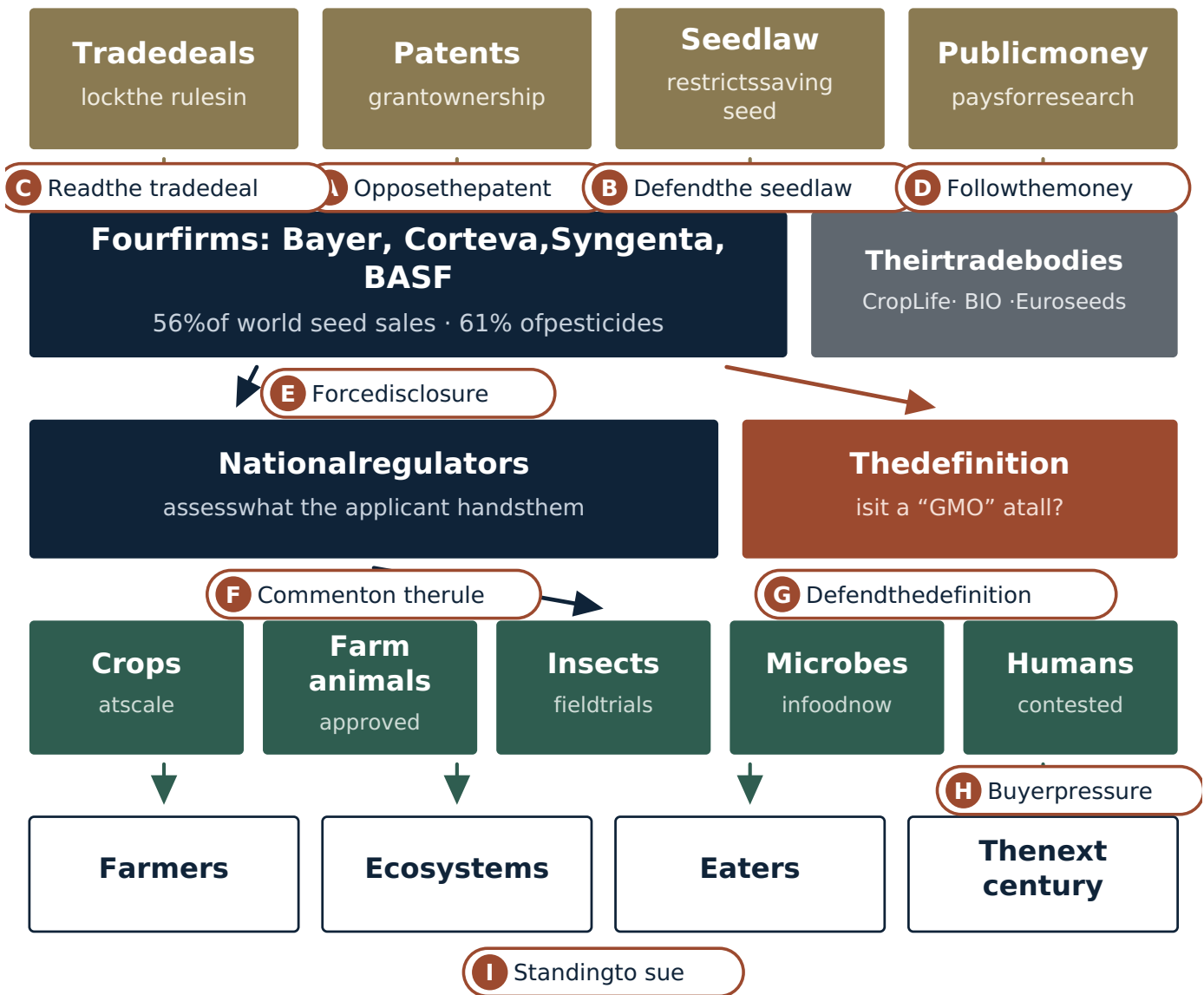


Figure 7. The same machine, with the nine handles marked. Each one attaches to a specific part of the structure, and each is described in the section that follows.

Nine handles. They are in rough order of how much you get back for the effort, not of importance. Every one of them is lawful, most are free, and several require nothing but a written submission before a date.

Before the list, three rules that decide whether any of it works.

Argue about process, ownership and consent — not about danger. You will not win a scientific safety argument, and attempting it costs you the arguments you would win. Who decided, who was asked, who profits, who pays when it goes wrong, and who was legally shut out are all questions where the record is on your side.

Go for the rule, not the product. A permit decides one product. A rule decides every product for a decade. Rule changes are decided on public dockets that receive, in most cases, a handful of substantive submissions.

Everything is lawful or it is worthless. Do not destroy crops, enter sites or damage equipment. Every documented instance of this has been used to convert the story from “the agency did not follow its own rules” into “these people are

vandals,” which is a story the other side wins. Your only real asset is that your file is accurate and your conduct is boring.

A. Oppose the patent

Cost: free to low. **Time:** months to years. **Who can:** anyone.

Most patent offices run a public opposition procedure with a fixed window after grant. At the European Patent Office it is nine months, and any person may file — no standing requirement, no lawyer required to start.

What to do: search the patent databases (Espacenet, Google Patents, WIPO Patentscope, all free) for patents on the crop, trait or organism you care about. Read the claims — the numbered list at the end is what is actually owned. Look for claims that reach beyond the specific engineered construct to plants merely containing a gene variant, including plants that arose by ordinary breeding.

Join an existing effort before starting your own. The No Patents on Seeds coalition has been running these oppositions for over a decade and will generally share its file.

Why this is the most under-used lever: patents last twenty years, apply across a whole region at once, and around 80% of the European public opposes patents on seeds. There is a large majority behind this position and almost nobody acting on it.

B. Defend the seed law

Cost: free to low. **Time:** ongoing. **Who can:** anyone, but this is a farmers’ fight and needs farmers at the front.

Who may save, exchange and sell seed is set by national plant variety legislation, and it is being changed continuously — usually toward UPOV 91 compliance, often as a condition attached to a trade agreement or a loan.

What to do: find out whether your country is a UPOV member, under which act, and whether a seed law amendment is in progress. Watch for regional harmonisation bills, which move several countries at once. Get farmers’ organisations, agricultural workers’ unions and seed-saver networks to the committee stage, which is where these bills are decided and where almost nobody appears.

Existing campaigns to join: the **STOP UPOV** campaign, the **Alliance for Food Sovereignty in Africa**, the **African Centre for Biodiversity**, **GRAIN**, **APBREBES**, and **La Via Campesina**. The **N’Djamena Declaration**, adopted at the fourth Pan-African Conference on Seed Governance in June 2026 with participants from twenty countries, is the current continental statement.

C. Read the trade deal

Cost: free. **Time:** whenever one is being negotiated. **Who can:** anyone who will read a long document.

Trade agreements lock rules in for decades and are negotiated with less public participation than almost any other instrument. They are where UPOV 91 compliance gets imposed, where biotechnology approval standards get harmonised downward, and where a future government’s freedom to change its mind gets signed away.

What to do: when your country is negotiating an agreement, find the intellectual property and the sanitary-and-phytosanitary chapters and read them for two things — any requirement to join or comply with UPOV 91, and any requirement to accept another party’s biotechnology approvals. Get those clauses in front of legislators before ratification. After ratification is too late.

Live examples at the time of writing: EU negotiations with the Philippines and with a group of south-eastern African states, both containing UPOV 91 clauses that farmers’ organisations in those countries are contesting; and the plant variety protection annex under the African Continental Free Trade Area.

D. Follow the public money

Cost: free. **Time:** weeks. **Who can:** anyone.

A large share of this research is paid for publicly or philanthropically, and those funders answer to bodies that will respond to a constituent.

What to do: search the grant databases of your national research council and agriculture department, and of the major

foundations. Note the project, the budget, and the commercial partner. Then ask the questions that funders find hardest: what are the terms of the licence on the resulting intellectual property, will the product be patented, and will farmers in the country that paid for it be able to save the seed?

Two additional channels people forget: most large funders have their own safeguards and complaints offices, separate from any regulator and far less used; and universities have institutional biosafety and research ethics committees whose minutes are often obtainable under freedom of information law at public institutions.

E. Force disclosure

Cost: free to low. **Time:** one to six months. **Who can:** anyone; most information laws are open to non-citizens.

The systematic weakness of this field is that the evidence is supplied by the applicant and much of it is withheld. Requests that work:

- The studies relied on for a specific approval, and who conducted and funded each one.
- The legal basis claimed for each redaction in a submitted file.
- The monitoring reports required as conditions of an earlier approval — frequently these do not exist, and the reply saying so is itself a story.
- Correspondence between the agency and the applicant over a defined period.
- Committee membership, declarations of interest, and minutes.

Ask narrowly and specifically. Broad requests get refused. Appeal refusals — most systems have an internal appeal and then an information commissioner, and refusals are overturned more often than people expect.

F. Comment on the rule

Cost: free. **Time:** hours to days, before a deadline. **Who can:** anyone, in most systems including non-residents.

This is the highest-value hour in the guide. Rule-making dockets on biotechnology typically receive a modest number of substantive submissions, overwhelmingly from industry. A well-made public-interest comment has to be answered on the record, and that answer then constrains the agency and gives a court something to review.

What to look for: any consultation on how engineered organisms are defined, classified, exempted, assessed or labelled. In the United States, the Federal Register and regulations.gov. In the EU, Commission “have your say” consultations and the comment window EFSA opens after publishing an opinion. In most other countries, the biosafety authority’s or environment ministry’s consultation page.

What a strong comment contains: the docket number; who you are and why you have an interest; a specific defect with a page or paragraph reference; what you want instead, stated as something the agency can actually do; and a fallback ask, so an agency unwilling to refuse can still impose a condition. Do not send a form letter — identical submissions are counted once.

G. Defend the definition

Cost: free. **Time:** ongoing. **Who can:** anyone.

This is F applied to the single question that matters most, and it deserves naming separately because it is where the field is actually being decided. If an organism is not legally a GMO, then no assessment, no notification, no comment window, no traceability and no label applies to it.

Arguments that have carried weight in this specific fight, none of which require claiming the organisms are dangerous:

- **Traceability and coexistence.** If a class of organisms is not tracked, an organic or non-GM producer cannot prove their product is what they say it is, and cannot defend a market that depends on it.
 - **The right to know.** Consumer choice arguments poll extremely well and are about information, not safety.
 - **Export exposure.** If your country deregulates a class its export markets still regulate, its own producers carry the risk.
 - **Reversibility.** A definition change is not product-by-product. It removes a whole category of future decisions from public view at once, which is a reason to require a higher standard of justification.
-

H. Reach the buyers

Cost: free. **Time:** months. **Who can:** anyone.

The market has no deadlines, no procedure and no obligation to say yes to anything. It is the one door that is never shut, and it is where several products actually died.

Monsanto shelved herbicide-tolerant wheat in 2004 after wheat buyers and growers' organisations made clear they would not accept it, while the approval process was still running. The NewLeaf engineered potato was withdrawn after major fast-food buyers declined it. Engineered salmon spent years struggling for shelf space because retailers publicly committed not to stock it.

Who to write to: grain traders, food manufacturers, supermarket chains, restaurant groups, and — often the most effective and most overlooked — the export trade bodies for your country's grain, honey, seed or organic sectors, who have a direct commercial interest and direct access to ministers. Their question is not "is it safe" but "will this cost me customers or shipments." Answer that question, in one page, with the export data attached.

I. Standing to sue

Cost: high without a free lawyer. **Time:** one to four years. **Who can:** usually organisations, sometimes individuals.

Courts almost never rule on whether an organism is safe. They rule on whether the agency followed its own rules. The winning grounds, repeatedly:

- The agency skipped a step its own law required.
- It did not consult when it was required to.
- Its decision was not supported by the evidence before it.
- It failed to apply the precautionary principle where its constitution or statute required it.
- Its decision-makers had conflicts of interest.

Precedents worth citing in your own filings:

- **India, 23 July 2024.** In the GM mustard litigation the Supreme Court split on the outcome but agreed on two points of principle: that decisions of the Genetic Engineering Appraisal Committee **are open to judicial review**, and that the government must **develop a national GM policy in consultation with states, farmers' groups and other stakeholders**, with proper verification of experts' credentials and mitigation of conflicts of interest. That is a usable precedent well beyond India.
- **United States, 2 December 2024.** A federal court vacated the SECURE rule. A rule that deregulates can be challenged on how it was made.
- **Philippines, 17 April 2024.** The Court of Appeals revoked biosafety permits for two crops — not on proof of danger, but because the government had not established the monitoring it was supposed to, applying the precautionary principle and placing the burden on the side seeking to change the status quo. Brought by a farmer-led network using the writ of *kalikasan*, a constitutional environmental remedy.
- **Mexico, 2013 onward.** A collective lawsuit froze GM maize planting permits for years, and the fight ended in a constitutional amendment.

Free or low-cost help exists and is under-asked: the Environmental Law Alliance Worldwide, ClientEarth, the Center for International Environmental Law, Earthjustice, AIDA, Natural Justice, and university environmental law clinics.

Beyond the nine: the international channels

Slow, mostly non-binding, free, and outside your government's control — which is exactly why they are worth using when the domestic system is closed.

The Cartagena Protocol and the Biosafety Clearing-House (bch.cbd.int). The free public database of national biosafety decisions, laws, risk assessments and each country's competent authority. If you do not know who decides in your country, you will know in five minutes. Article 26 permits countries to take socio-economic considerations into account, including effects on indigenous and local communities. Most ignore it. It is available to be asked about.

The Convention on Biological Diversity. Synthetic biology and gene drives are handled at its Conferences of the Parties, which accept written submissions from civil society. **COP17 runs 19–30 October 2026 in Yerevan, Armenia**, alongside

the twelfth meeting of the Cartagena Protocol parties and the sixth of the Nagoya Protocol parties. Synthetic biology and digital sequence information are both on the agenda. This is the single largest scheduled opening in the next year.

The Aarhus Convention Compliance Committee. Genuinely unusual and badly under-used: **any member of the public may bring a communication** alleging that a party has breached the Convention’s rights to information, participation and justice in environmental matters — free, no lawyer, no need to be a national. The Convention has a specific GMO amendment on public participation. If your government ran an eleven-day consultation in one language on a website nobody affected reads, this is the body that exists for that complaint.

UN Special Rapporteurs on the right to food, on toxics and human rights, and on the rights of indigenous peoples. A short factual submission by email. They cannot order anything; they can write to your government and publish it.

Free, prior and informed consent. Where a release or a seed law affects indigenous territory, this is a recognised standard with binding regional human rights judgments behind it. Nothing else in this field carries that weight. If this is your situation, it belongs at the centre of everything you file.

OECD National Contact Points, for conduct by a multinational headquartered in an OECD country.

PART SIX: THE HONEST SCORECARD

Before committing a year to any of this. These are judgements about effort against effect, drawn from how documented campaigns have gone. **They are not measured data and no controlled study of this exists.** Treat them as a ranking, not a prediction.

“Works” means the rule is changed, delayed, conditioned, or an obligation is created that did not exist before.

What you do	Rough odds it changes something	Time	Cost
Read the docket and do nothing else	near zero	days	none
One substantive comment on a rule-making	10-25%	hours	none
Same, with 20+ affected signatories and two named experts	25-45%	1-3 months	none
Freedom of information request for the withheld studies	20-40%	1-6 months	none
Patent opposition, joining an existing coalition	20-40%	1-3 years	low
Getting UPOV clauses out of a trade deal before ratification	10-25%	months	low
Public-funder accountability question via a legislator	15-35%	months	none
Buyer, exporter or retailer commitment	20-45%	months	none
Aarhus or Special Rapporteur complaint	10-25%	1-2 years	none
Written submission to a treaty COP	5-20%	months	low
Strategic litigation on process	20-40%	1-4 years	high without a free lawyer
Legislation or constitutional change	5-15%	3-15 years	high
All of it, run together, sustained	35-60%	3-10 years	low to moderate

Three things that table is trying to say.

The cheapest row is the most skipped. A twelve-page comment on a rule-making docket, with paragraph references and twenty signatures from people with a concrete interest, is a different object from the one-line objections agencies dismiss in a sentence. It has to be answered on the record, and that answer is then something a court can read.

Nothing in the middle rows works alone. The comment makes the story printable. The story makes the legislator answer.

The legislator's question makes the funder respond. The funder's response is evidence in the next comment.

The bottom row takes years and is where the durable wins are. India's Bt brinjal moratorium has held since 2010. Mexico's maize prohibition is in the constitution. Both took over a decade.

PART SEVEN: WHERE TO GO, COUNTRY BY COUNTRY

Every country has a version of the same four bodies: a biosafety regulator, a patent office, a plant variety office, and a legislature. This section gives the specific ones for the countries where most of the deciding happens, then the general route for everywhere else.

The global instruments — start here whoever you are

What	Where	What it gives you
Biosafety Clearing-House	bch.cbd.int	National decisions, laws, risk assessments and the competent authority for every Cartagena party
CBD COP17 / Cartagena MOP12	Yerevan, 19–30 Oct 2026	Synthetic biology and gene drives on the agenda; accepts written submissions
Aarhus Compliance Committee	unece.org	Any member of the public may file a communication, free
UN Special Rapporteurs	ohchr.org	Right to food; toxics; indigenous rights. Email submissions
UPOV	upov.int	Which act your country is bound by
Plant Treaty (ITPGRFA)	fao.org/plant-treaty	Farmers' rights language for seed-law arguments
Espacenet / Patentscope	free	What is actually patented, and its claims

United States

The largest producer and the loosest framework. Three agencies under the 1986 Coordinated Framework: **APHIS** (agriculture department; plants, under authority tied to plant pests), **EPA** (anything engineered to act as a pesticide), **FDA** (food and animals).

Current position: the 2020 SECURE rule was vacated on 2 December 2024 and the agency reverted to the pre-2020 regulations. It has indicated it will no longer perform environmental analyses under the National Environmental Policy Act for deregulation petitions. A **Request for Information on the future framework was published 15 May 2026**, with comments closing 15 June 2026, and an interim final rule listed on the regulatory agenda for 2026. The replacement framework is being written now.

Where to act: **regulations.gov** for comments, **Federal Register** for notices, the APHIS permit and notification data files for what is actually being released, USDA NIFA's grant database for public funding, the Senate and House lobbying disclosure databases, and USPTO plus PTAB for patents.

European Union

Historically the strictest framework and currently the most significant single change. **Regulation (EU) 2026/1388** on new genomic techniques entered into force 16 July 2026 and applies from 17 July 2028, splitting gene-edited plants into a lighter "NGT-1" tier and an "NGT-2" tier closer to the old rules. The Parliament's proposed ban on patents for such plants is not in the final text.

Two consequences. The implementing acts, technical criteria and verification procedures are being written between now and 2028, which is a long open rule-making window. And the patent question was left to patent law, which means the European Patent Office opposition route is where it now has to be fought.

Where to act: EFSA's comment window (about 30 days after it publishes an opinion; applications via Connect.EFSA, documents on OpenEFSA), Commission "have your say" consultations, the EU Transparency Register and LobbyFacts for lobbying, the European Ombudsman for maladministration, the CJEU, and the EPO opposition procedure.

Brazil and Argentina

Between them, close to 90 million hectares. **Argentina** was the first country in the world to place gene-edited crops outside GMO rules, in 2015 — the precedent everyone else cites — through CONABIA. **Brazil** followed in 2018 through CTNBio, whose decisions and meeting records are published.

Where the argument is live: soy expansion and deforestation, pesticide exposure in rural communities and around schools (Córdoba and Entre Ríos have produced convictions and buffer rules), and seed royalty systems. Argentina's provinces and Brazil's states have real regulatory power here.

India

Regulated by process, through the **Genetic Engineering Appraisal Committee** under the environment ministry, whose meeting agendas and minutes are published. Bt cotton is the only commercially cultivated GM crop; the **Bt brinjal moratorium of February 2010 has never been lifted**.

The live position: the Supreme Court's split verdict of 23 July 2024 confirmed GEAC decisions are judicially reviewable and directed the government to formulate a **national GM policy in consultation with states, farmers' groups and other stakeholders**, with verification of experts' credentials and mitigation of conflicts of interest. That consultation is the opening. Gene editing without foreign DNA has been given a lighter path, and imports of GM soy and maize for feed are under active political argument.

China

Now decisive and the hardest to work in. Biotechnology is treated as a national security question. Engineered maize and soy expanded four to five fold in 2025, to roughly three million hectares; the first gene-edited rice was approved for cultivation in December 2024. Reporting indicates that **from 2025 the agriculture ministry stopped publishing newly approved biosafety certificates**, which is itself the most useful thing to raise.

For those outside China, the practical routes are trade and import: what Chinese approvals mean for exporters, and what Chinese imports mean for producing countries.

Canada and Australia

Both matter because of what they permit. **Canada** regulates by novelty of trait rather than by technique, so gene-edited and transgenic products sit in the same register — its **CFIA "Plants with Novel Traits"** dataset is one of the few places the deregulated class remains visible. **Australia's** Office of the Gene Technology Regulator publishes a risk assessment and management plan for each licence with a minimum 30-day comment period, advertised in the Government Notices Gazette with a free alert list, and it publishes the locations of active field trial sites — almost uniquely.

Australia's statutory limitation is worth knowing before you write: the regulator is **barred from considering trade, marketing, food safety or the benefits of a product**. Arguments outside that scope must go to parliament, the press or the market instead.

United Kingdom

The Genetic Technology (Precision Breeding) Act 2023 created a separate, lighter regime for "precision bred" plants and animals in England, distinct from the rest of the UK — which makes internal-market and devolution arguments available that do not exist elsewhere. Scotland and Wales have taken different positions.

Africa

The fastest-moving region and the one where seed law matters most. Nigeria, Kenya, Ghana, Malawi, Ethiopia and Burkina Faso have issued genome-editing guidance, and dozens of projects are running across the continent. South Africa regulates all gene-edited crops as GM. Burkina Faso has now ended two flagship programmes on its own terms — Bt cotton, and the engineered mosquito work.

The seed-law front: seventeen West African countries are bound to UPOV 91 through OAPI membership; Kenya's and Ghana's seed laws carry criminal penalties; the East African Community's seed bill and the AfCFTA plant variety annex would move many countries at once.

Organisations to work through: the **Alliance for Food Sovereignty in Africa**, the **African Centre for Biodiversity**, and the regional farmer networks. The **N'Djamena Declaration** of June 2026 is the current continental position.

Latin America beyond Brazil and Argentina

Mexico is the most important case in the world for centre-of-origin arguments and now has a constitutional prohibition on

planting GM maize. Chile, Colombia, Paraguay, Honduras, Guatemala, Costa Rica and Uruguay have all adopted Argentina-style gene-editing exemptions. **Peru** has a moratorium.

The **Escazú Agreement** gives Latin American and Caribbean parties an environmental information, participation and justice instrument comparable to Aarhus, with specific protections for environmental defenders.

Everywhere else

The route is the same in every country and takes about an evening:

1. Look your country up on **bch.cbd.int** to find the competent national authority and the national biosafety law.
2. Find whether it is a **UPOV** member and under which act.
3. Find the national **patent office** and its opposition procedure.
4. Find the **freedom of information** law and whether non-citizens may use it.
5. Find whether it is a party to **Aarhus** or **Escazú**.
6. Find which **regional body** — OAPI, ARIPO, COMESA, the African Union, ASEAN, Mercosur — makes rules that bind it without a separate national decision.

That is the map. Nearly everything in Part Five attaches to one of those six.

Organisations working on this internationally

Not endorsements, and their positions differ. These are the groups that hold files, run the existing campaigns, and will usually share what they have:

Seed and patents: No Patents on Seeds, APBEBES, GRAIN, Arche Noah, the Corner House. **Food sovereignty and farmers:** La Vía Campesina, the Alliance for Food Sovereignty in Africa, MASIPAG, the African Centre for Biodiversity, Third World Network. **Biotechnology watchdogs:** ETC Group, GeneWatch UK, GMWatch, the Canadian Biotechnology Action Network, Testbiotech, the Center for Food Safety. **Corporate and lobbying research:** Corporate Europe Observatory, LobbyFacts, DeSmog, LittleSis. **Legal:** the Environmental Law Alliance Worldwide, ClientEarth, the Center for International Environmental Law, Earthjustice, AIDA, Natural Justice.

PART EIGHT: LETTERS THAT DO STRUCTURAL WORK

Each of these asks for something and creates a record that you asked. Keep copies. Send anything important by a method that proves it arrived. Log everything with dates.

Replace anything in square brackets. Keep them short and attach your evidence rather than describing it.

TEMPLATE A — Comment on a rule-making or consultation

To: [the agency] · **Subject:** Comment on [docket / consultation reference] — [short title of the proposal]

I am writing about [reference], on which comments close on [date].

I am [who you are and what interest you have — “a certified organic grower”, “a seed-saving network with 240 members”, “an exporter to [market]”].

1. What the proposal does. As drafted, [paragraph N] would [state the effect plainly — for example, place organisms produced by [technique] outside the scope of these regulations].

2. The consequence not addressed. [One specific, checkable consequence. For example: producers certified to a standard that excludes these organisms would have no means of verifying compliance, because nothing in the proposal requires the organisms to be identifiable.]

3. What is missing from the record. The proposal does not [assess the effect on export markets that continue to regulate this class / explain how traceability is preserved / identify who bears the cost when it is not].

I ask the agency to: (a) [the primary ask]; (b) failing that, [a fallback the agency can grant without refusing outright — a notification requirement, an identifier requirement, a review clause, a reporting obligation]; and (c) place on the record its response to each point above.

[Name, address, date. List of co-signatories and their interest.]

TEMPLATE B — Freedom of information request

To: [information officer] · **Subject:** Request under [name of the law]

Under [the law], I request:

1. The studies relied on in reaching [decision / approval reference], together with, for each, the identity of the party that conducted it and the party that funded it.
2. For each redaction applied to [document], the specific statutory exemption relied upon.
3. The monitoring reports submitted under condition [N] of [approval] for [years].
4. The membership of [committee], with declarations of interest, and minutes of the meeting of [date].

If any part is withheld, please state the exemption for each document and release the remainder. Please confirm receipt and the date by which you expect to reply.

[Name, contact, date]

TEMPLATE C — Question to a legislator about public funding

To: [representative] · **Subject:** Publicly funded research on [subject] and what happens to the results

I am a constituent in [place].

[Funder] has awarded [amount / project reference] to [institution] for [project], in partnership with [company]. The grant record is at [link].

I would like to ask, through you:

1. What are the terms of the intellectual property licence arising from this public funding? Is the resulting material patented, and by whom?
2. Will farmers in [country] be able to save and replant seed derived from it?
3. What independent assessment, not funded by the commercial partner, has been commissioned?
4. Which official is accountable for the answer?

These are questions of public expenditure rather than of science, and I believe they are properly asked of [ministry].

[Name, address, date]

TEMPLATE D — Submission on a trade agreement

To: [trade ministry / parliamentary committee] · **Subject:** [Agreement] — intellectual property chapter, plant variety provisions

We are [who]. We write regarding [Article N] of the draft [agreement], which would require [country] to [accede to UPOV 91 / bring its plant variety law into conformity with UPOV 91].

What this changes domestically. [State plainly: under current national law, farmers may [do X]. Under a UPOV 91-compliant regime, [that becomes subject to the breeder's authorisation / an offence].]

Who is affected. [Number of smallholders, share of seed supplied by farmer-managed systems, the crops concerned.]

What we ask. That [Article N] be removed, or that the agreement instead require an effective sui generis system consistent with the International Treaty on Plant Genetic Resources for Food and Agriculture, which [country] has ratified, and with the right to seeds recognised in UNDROP.

We ask that this be considered before ratification, since the obligation cannot be revisited afterwards without reopening the agreement.

[Names, organisations, date]

TEMPLATE E — Patent opposition enquiry

To: [No Patents on Seeds / a patent attorney / a law clinic] · **Subject:** Possible opposition to [patent number]

Patent [number], granted to [holder] on [date], claims [summarise the claims in one or two sentences — particularly any claim reaching beyond a specific engineered construct to plants merely containing a gene variant].

Our concern is that the claims extend to [plants obtained by conventional breeding / naturally occurring gene variants], which we understand to be excluded from patentability under [provision].

The opposition period closes on [date]. We can contribute [evidence of prior use / breeding records / affected breeders willing to give statements / funds].

Can we discuss before the deadline?

[Name, contact, date]

TEMPLATE F — Communication to the Aarhus Compliance Committee

To: the Aarhus Convention Compliance Committee, UNECE · **Subject:** Communication concerning compliance by [Party]

[Use the Committee's required format, available on the UNECE site.]

The facts. On [date], [authority] adopted [decision / regulation] concerning [subject]. Notice was published on [date], in [language(s)], for [N] days. [State who was affected and how they learned of it.]

The provisions engaged. Article 6 and Annex I bis (public participation in decisions on the deliberate release of genetically modified organisms); Article 4 (access to information) in respect of [the withheld material]; Article 9 (access to justice) in respect of [the absence of a review route].

Domestic remedies. [What was attempted and what happened, or why no effective remedy exists.]

What we ask. That the Committee find [Party] not in compliance with [articles] and recommend [specific measures].

Evidence attached: [list].

TEMPLATE G — Letter to a buyer or export body

To: [buyer / trade body] · **Subject:** Traceability risk from the proposed reclassification of [category] in [country]

We are writing about [proposal reference], which would place [category] of organisms outside [country]'s [GMO / biotechnology] rules.

[Market] continues to regulate this class and requires [documentation / threshold compliance]. If these organisms are not identifiable in [country], a supplier cannot demonstrate compliance for shipments into that market.

The precedent is [StarLink maize in 2000 / LibertyLink rice in 2006], where contamination closed export markets and produced litigation lasting years.

We ask you to state publicly, before [date], that you require [identifiable supply / continued traceability], and to write to [ministry] setting out the commercial exposure.

[Names, contact, date]

TEMPLATE H — Submission to a treaty meeting

To: [CBD Secretariat / the relevant subsidiary body] · **Subject:** Submission of views on [agenda item] ahead of [meeting]

[Organisation] submits the following views on [item — e.g. synthetic biology; risk assessment of living modified organisms containing engineered gene drives].

Observation. [One or two paragraphs of specific, sourced fact — what has actually happened in your country or region, which nobody in the room will otherwise report.]

Concern. [The governance gap this reveals, in the treaty's own language.]

Suggested text. [A concrete sentence or clause Parties could adopt. Be specific; general concern is discarded and drafted text sometimes survives.]

[Organisation, contact, date]

IF YOU HAVE ONLY A FEW HOURS A WEEK

In order of what you get back for the time:

1. **Find out whether a rule change is open in your country right now.** One evening. Search your biosafety authority and environment ministry for consultations, and your national gazette or register for notices. If one is open, everything else waits.
2. **File one substantive comment before the deadline.** One evening, using Template A. This is the highest-value hour in the guide.
3. **Get three affected people to file their own, in their own words.** A week of phone calls. Identical letters count once; different ones do not.
4. **Send one freedom of information request.** Thirty minutes, Template B. It works while you sleep, and a refusal is itself usable.

5. **Look up your country on the Biosafety Clearing-House and on UPOV.** An hour. You now know who decides and under which seed regime.
6. **Ask one legislator one funding question.** An hour, Template C.
7. **Join an existing coalition rather than starting one.** They have the file already.

Do only the first three and you have put a specific, answerable objection on the official record before the decision was made, which is more than most opposition to this industry has ever managed.

GLOSSARY

Aarhus Convention — a European treaty on environmental information, participation and justice, with a compliance committee open to any member of the public and a specific amendment about GMO decisions.

Biosafety Clearing-House — the free public database at bch.cbd.int holding national biosafety decisions, laws and deciding authorities for every Cartagena party.

Bt — short for *Bacillus thuringiensis*, a soil bacterium. Bt crops are engineered to make one of its insecticidal proteins.

Cartagena Protocol — the international agreement on moving living modified organisms between countries. In force 2003, 173 parties. The US is not one.

Centre of origin — the region where a crop was first domesticated and where its traditional varieties and wild relatives still exist. Maize in Mexico and Central America, potato in the Andes, rice in Asia.

Coordinated Framework — the 1986 US decision to regulate biotechnology under existing statutes rather than pass a new law. Still the basis today.

CRISPR — the best-known gene-editing tool. Cheap, fast, widely available.

Escazú Agreement — the Latin American and Caribbean equivalent of Aarhus, with specific protections for environmental defenders.

Free, prior and informed consent (FPIC) — the standard that indigenous peoples must consent, in advance and fully informed, to activities affecting their lands. Recognised in UNDRIP and enforced in regional human rights courts.

Gene drive — an engineered system built to spread a trait through a wild population faster than ordinary inheritance. Self-spreading and, in practice, not recallable.

GMO / LMO — genetically modified organism; living modified organism, the term used by the Cartagena Protocol.

Herbicide tolerance — engineering that lets a crop survive a weedkiller that would otherwise kill it. The most widespread engineered trait by area.

NGT — “new genomic techniques”, the EU’s term for gene editing, now split into NGT-1 and NGT-2 tiers by Regulation (EU) 2026/1388.

OECD unique identifier — the code that lets a specific engineered event be tracked across countries. Designed around transgenic insertions, so gene-edited organisms without an inserted transgene generally do not get one.

Precautionary principle — that a lack of full scientific certainty is not a reason to postpone measures against serious or irreversible harm. Written into many national laws and constitutions.

SDN-1, SDN-2, SDN-3 — categories of gene edit. Roughly: a cut with no template, a cut with a small template, and an insertion of foreign DNA. Many countries regulate only the third as a GMO. This distinction is doing most of the legal work worldwide.

Sui generis — “of its own kind”. A plant variety protection system a country designs for itself, permitted by WTO rules, as an alternative to UPOV 91.

UNDROP — the UN Declaration on the Rights of Peasants, which recognises a right to seeds.

UPOV — the treaty on plant variety protection. The 1991 act sharply narrows farmers’ rights to save and exchange seed. 80 members covering 99 countries as of mid-2025.

YOUR FIRST MONTH: A CHECKLIST

- ☐ Look your country up on bch.cbd.int. Write down the competent authority.
 - ☐ Find your national biosafety law and when it was last amended.
 - ☐ Check whether your country is a UPOV member and under which act.
 - ☐ Check whether it is a party to Aarhus or Escazú.
 - ☐ Find the regional body that makes rules binding on it.
 - ☐ Search for any open consultation on definitions, classification or exemptions. Write the closing date somewhere you see it daily.
 - ☐ Read the actual proposal, not the summary. Note three things it does not address.
 - ☐ Search a free patent database for patents on the crop or trait you care about. Read the claims.
 - ☐ Search your national research funder's grant database for public money in this area.
 - ☐ Look up the four firms in your national lobbying register.
 - ☐ Find the two or three organisations already working on this in your country and ask what they need.
 - ☐ Find two other people who will still be doing this in eight months.
 - ☐ File one comment. File one information request.
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This document describes lawful methods only: reading public records, requesting information, filing comments, oppositions and appeals, organising people, writing to legislators and buyers, and using courts and international bodies. It is not legal advice. Rules, deadlines and authorities differ by country and change often, and several of the frameworks described here are being actively rewritten. Verify every procedural detail against your own jurisdiction's current requirements before relying on it.